Methodological Falsifiability and Dependency Integrity Standard v2.0

Source-boundary rule: If this compact standard conflicts with the RippleLogic Canon, the Canon controls.

Release: MathGov Core Release 2026.09 v12.3 SGP v8.2 Reality-Management Calibration Integration Public Research Source Release

Purpose

This compact standard defines the v12.0 methodological integrity discipline for MathGov and RippleLogic. It adopts the bounded principle that claim-bearing elements must not survive merely because they are plausible, fluent, approved, compliant, or locally useful. A material claim must declare what kind of claim it is, what it depends on, what would force revision, what alternatives remain open, and what downstream elements must be rerun if the foundation changes.

It is not a sixth gate. It does not replace Reality Grounding, RF/NCRC, TRC, CSV, SGP, RLS, PC-AEP, domain science, or external validation.

Adopted principles and bounded non-claims

Item	MathGov integration
Adopted principle	Material claims need external contact, falsification or revision triggers, explicit dependencies, and no hidden assumption authority.
Adopted principle	If a starting assumption, definition, threshold, operator, or model basis changes, the affected downstream chain must be rerun or claim strength must be narrowed.
Bounded non-claim	Not every normative definition is physically falsifiable. Normative commitments are audited for coherence, rights compatibility, stakeholder challenge, and application integrity.
Bounded non-claim	Probabilistic evidence is not invalid. It is required where uncertainty is real, but uncertainty must be disclosed and cannot be laundered into certainty.
Bounded non-claim	Model revision is not forbidden. Silent retrofitting is forbidden. Revision must be versioned, disclosed, and rechecked.

Required Methodological Integrity Record

Field	Required content
claim_or_component	The claim, definition, threshold, operator, indicator, model, SGP interpretation, gate decision, validation result, or public release claim under review.
claim_type	empirical, causal, formal, operational, normative, or metaphysical-horizon.
dependency_position	Where the claim sits in the cascade, standard, SGP flow, workbook, validation protocol, or public release surface.
starting_assumptions	Assumptions, definitions, priors, sources, thresholds, scope boundaries, or model premises that carry the claim.

Field	Required content
definition_or_operator_used	The material definition, operator, threshold, indicator, or procedure being applied.
necessity_or_alternative_check	Whether necessity is claimed, what alternatives remain plausible, and why the stronger claim is or is not warranted.
evidence_or_test_surface	The observation, measurement, replication, derivation, simulation, formal proof, expert review, stakeholder challenge, or audit surface used.
falsification_or_revision_trigger	What result, evidence, challenge, failure, or domain change would force revision, narrowing, escalation, or refusal.
uncertainty_or_confidence_method	How uncertainty, confidence, model limits, disagreement, or unresolved evidence is represented.
downstream_dependencies	Gates, scores, SGP outputs, claims, controls, authority actions, or release statements affected if this element changes.
re_derivation_scope_if_changed	Local patch, threshold update, gate rerun, TRC reopening, CSV rerun, SGP rerun, RLS recalculation, release versioning, or full refusal.
reviewer_status	reviewer-visible, contested, independently checked, domain-expert-warranted, or pending.
required_claim_action	proceed, narrow, control, redesign, escalate, or refuse.

Status values

Status	Meaning	Default action
METHOD_SUPPORTED	Evidence, dependency, and revision conditions support the declared claim boundary.	Proceed within boundary.
ASSUMPTION_BOUND	Claim depends on visible assumptions that could affect strength or gate status.	Narrow or monitor.
ALT_EXPLANATION_OPEN	Plausible alternatives can explain the same result.	Downgrade necessity claim.
TEST_REQUIRED	Stronger evidence, derivation, replication, simulation, or review is needed.	Escalate before strong claim.
CONTESTED	Evidence, interpretation, stakeholder challenge, or expert judgment materially disputes the claim.	Escalate, narrow, or redesign.
REVISION_REQUIRED	A starting assumption, definition, threshold, operator, or model basis has failed or changed.	Rerun affected chain.
REFUSE_STRONGER_CLAIM	The requested public, deployment, validation, deterministic-selection, or safety claim exceeds support.	Refuse the stronger claim.

Claim-type rules

Claim type	Evidence discipline
Empirical	Observation, measurement, replication, current source, or empirical warrant.
Causal	Mechanism, intervention, counterfactual, domain model, or qualified causal warrant.
Formal	Explicit assumptions, definitions, derivation, proof, or formal verification.
Operational	Reproducible procedure, audit trail, role assignment, and reviewable outputs.
Normative	Declared ethical commitment, rights compatibility, consistency, stakeholder challenge, and auditability of application.
Metaphysical-horizon	Humility and meaning orientation only; no Tier 1-3 evidence, gate override, or authority by itself.

Non-retrofitting rule

A failed test, falsified prediction, changed foundation, or challenged dependency must not be hidden by silent tuning. The run must version the change, record the affected dependencies, and rerun every materially dependent gate, SGP interpretation, calculation, authority action, or public claim.

Proportionality rule

Methodological rigor is proportional to stakes. Tier 1 and low-stakes Tier 2 uses may use lighter records. Tier 3, public conformance, institutional deployment, high-stakes SGP use, validation claims, and physical/causal actions require stronger records.

Physical-admissibility claim discipline

A claim that a physical execution is safe, admissible, stable, reliable, or deployment-ready is a material claim and must name its external evidence surface. Governance permission, compliance status, documentation, certification, monitoring, and authority are not sufficient evidence surfaces by themselves. They may support procedural legitimacy, but they do not compute physics or establish biological, mechanical, medical, infrastructural, or cyber-physical safety.

Computational-source discipline. A material execution claim SHOULD distinguish the source that generated the candidate action from the source that warrants the action. If candidate generation and admissibility warrant are supplied by the same system, the claim record SHOULD state how circular self-confirmation is prevented and what independent check, validity-domain limit, or refusal condition applies.

If a physical-admissibility claim changes its model, validity domain, threshold, controller envelope, failure-mode account, or monitoring/shutoff assumptions, affected downstream gates must be rerun. Silent tuning may improve wording, but it does not preserve the original claim.

Claim-type examples

Claim type	Example	Minimum discipline
Empirical	“Injury rates decreased after the intervention.”	Requires evidence contact, baseline, uncertainty, and alternative-explanation review.
Causal	“The policy caused the decrease.”	Requires mechanism, counterfactual, intervention logic, or qualified causal warrant.

Claim type	Example	Minimum discipline
Formal	“CVaR was computed by the declared algorithm.”	Requires definitions, assumptions, formula, inputs, and reproducible calculation surface.
Operational	“The workflow can be executed by this agency.”	Requires procedure, role owner, resource evidence, audit trail, and failure handling.
Normative	“This rights floor is adopted as non-compensatory.”	Requires declared ethical commitment, consistency check, rights compatibility, and challenge pathway.
Metaphysical horizon	“AIU language is used as a humility horizon.”	May guide meaning and restraint, but creates no Tier 1-3 evidence or gate authority.